

Sadguru Balumama Shikshan Prasarak Mandal's
K.P.PATIL INSTITUTE OF TECHNOLOGY

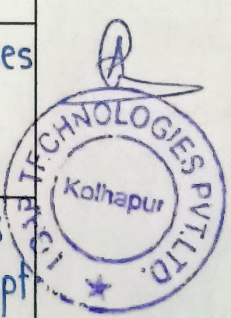
(DTE CODE : 6814) (MSBTE Code : 1661)

Approved by AICTE, DTE Mumbai & Govt. of Maharashtra, Affiliated to MSBTE Mumbai

Internships Daily Diary(2025-26)

Department Of Artificial Intelligence & Machine Learning Engineering

Program Code : AI5K
Name of the Student : Sanika Dhanaji chougale
Name of the Mentor (Faculty) : Mrs. P.M. Kumbhar
Name of the Mentor (Industry): Mr. Abhijit Gatade
Enrollment Number : 23213500044

Week	Day & Date	Discussion Topics/Activity	Details of Work Allotted Till Next Session /Corrections Suggested/Faculty Remarks	Signature of Industry Mentor
Week <u>5</u>	Mon, Date <u>30/06/25</u>	◆ Dictionaries and Sets	worked with key Value pairs, unique data	
	Tue, Date <u>01/07/25</u>	◆ Tuples & its functions	Learned about immutable data structures.	
	Wed, Date <u>02/07/25</u>	◆ file handling	Read /Write files using various modes	
	Thu, Date <u>03/07/25</u>	◆ Exception Handling	Managed errors using try-except blocks.	
	Fri, Date <u>04/07/25</u>	◆ Object-Oriented programming	Created classes and organized with OOP.	
	Sat, Date <u>05/07/25</u>	◆ Modules & Packages	Reused Code Using modules and packages	



Internships Daily Diary (2025-26)

Week No: 05

Day: Monday

Date: 30/06/25

Topic - "Dictionaries and Sets"

Today, I learned about two important data structures in python: dictionaries and sets.

Dictionaries store data in the form of key-value pairs, which allows fast look-ups and easy data mapping. I practiced Creating dictionaries, accessing and modifying elements using keys, and applying methods such as, `get()`, `pop()`, and `items()`.

Sets, on the other hand, are used to store unique and unordered data. I learned how to create sets and perform operations such as union, intersection, difference and Subset checking.

These data structure are especially useful when working with large, unorganized data.



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Internships Daily Diary (2025-26)

Week No: 05

Day: Tuesday

Date: 01/07/25

Topic - 'Tuples and Their Functions'

Today's session focused on tuples, an immutable data structure in python. I learned how tuples differ from lists and why they are preferred in cases where data integrity is crucial.

Tuples consume less memory & processing time compared to lists.

I also explored functions and methods applicable to tuples, such as `len()`, `index()`, and `count()`. I understood how tuples are often used in data grouping, as keys in dictionaries, and when returning multiple values from a function.

The session helped reinforce the importance of immutability in certain applications.



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Internships Daily Diary (2025-26)

Week No: 05

Day: Wednesday

Date: 02/07/25

Topic - "File Handling in Python"

Today, I studied how to handle files in Python for reading and writing purposes. I practiced opening files using the `open()` function in different modes ('r', 'w', 'a', 'r+') and understood how each mode behaves. I also explored how to read data using `read()`, `readline()` and how to write data using `write()` and `writelines()`.

I learned about the importance of using `with open()` Context manager to ensure that files are properly closed after operations which helps in avoiding memory leaks or file corruption.

This topic is essential for working with real-time data stored in external files.



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Internships Daily Diary (2025-26)

Week No: 05

Day: Thursday

Date: 03/07/25

Topic - "Exception Handling"

The focus of today's session was on error and exception handling in python. I learned how to identify and manage different types of runtime errors using try, except, else and finally blocks.

This technique allows a program to continue running smoothly even when an unexpected error occurs.

I also practiced writing code that raises exceptions using the raise keyword and explored built-in exception types like TypeError and FileNotFoundError. By implementing proper exception handling, I now understand how to build robust and fault.



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Internships Daily Diary (2025-26)

Week No: 05

Day: Friday

Date: 04/07/25

Topic - "Object - Oriented Programming"

Today, I was introduced to the fundamentals of Object - Oriented programming in python. I learned how to define classes and Create Objects, and how to use Constructors (—init—), instance Variables, and methods to encapsulate data and behaviour.

The session also Covered key OOP principles Such as inheritance, encapsulation, and polymorphism.

I Created a basic project where where multiple classes were organized and inherited from one another.

✓ This helped me understand how oop allows better Code Organization, reusability Software System.



Internships Daily Diary (2025-26)

Week No: 05

Day: Saturday

Date: 05/07/25

Topic - "Modules and Packages"

The final session of the week was about organizing python code using modules and packages. I learned how to create my own python module and how to import functions or classes from it into another Script using the import and from keywords.

Additionally, I was introduced to packages, which are collections of modules organized in directories with an `__init__.py` file.

This structure helps in building large - scale application with multiple modules.

This week helped my understanding Concepts. By working on real-time example and handling data in structured way.